Paleomagnetism of Annually Banded Eocene Green River Sediments

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A continuous set of samples from the Green River formation of Eocene age has been studied paleomagnetically. Remanent magnetization is found to be about 1–5 $\times$ 10^{-8} emu/cm^3 after demagnetization at 100 oe and is probably carried by the less than 0.002% magnetite present. The level of magnetization is exceedingly low, and it was necessary to take a large number of precautions in making the measurements. These samples are varved and contain layers about 0.18 mm thick, each couplet of which is believed to represent an annual depositional cycle. On this basis we have examined about 10,000 years of the record in roughly 100-year units. During most of this time the declination of magnetization is consistently nearly 90° away from the field presumed to represent the Tertiary in North America. This suggests that at least locally there can be a significant and consistent discrepancy between the rotation pole and the virtual magnetic pole.

The Green River formation is a formation occurring extensively in Utah, Colorado, and Wyoming and represents a large reservoir of untapped oil shale. As a result, it has been extensively studied and documented, largely in reports by *Bradley* [1928, 1931, 1964] and more recently by others [*Cashion*, 1967; *Donnell*, 1961]. It is a banded formation believed by many to represent annual varves formed in a lacustrine environment. The formation is extensively fossiliferous, and it is well documented as Eocene in age. Locally interbedded ash flows have been identified and dated. *Evernden et al.* [1964] have given dates ranging from 45 to 49 m.y. for some of these beds, confirming the Eocene age.

Complete stratigraphic sequences are not available for study, but it has been estimated that the formation may be as much as 4100 meters thick, although *Donnell* [1961] gives an approximate thickness of 1000 meters. *Bradley* [1928] has measured many varved sequences from the Green River formation under the microscope and gives a weighted mean of 0.18 mm for the thickness of each varve pair. We have not determined the thickness of varves from our samples, but a rough examination shows that 0.18 mm per varve pair is approximately correct for our section of samples. If this varve thickness holds true for the whole Green River sequence, it spans a period of about 5 m.y., which is close to the 4-m.y. life span quoted by *Bradley and Eugster* [1969]. We have collected a small sequence of samples from the Parachute Creek member of the formation near the mouth of Piceance Creek in Rio Blanco county (T IN, R97W) in Colorado (Figures 1*a* and 1*b*). A continuous collection representing about 2.7 meters was taken from a freshly blasted surface in an area under intensive study by the U.S. Bureau of Mines.

The samples were cut into 89 2.54-cm lengths for measurement on a PAR SM-2D spinner magnetometer. The cores cut from the field collection represent a nearly continuous sequence with only occasional small gaps and discontinuities in the 2.7-meter section. This means that, on the average, each 2.54-cm core represents approximately 140 years. This value is only an estimate, but it is not likely to be in error by more than a factor of 2. It means that only features of the magnetic field that have fluctuations with periods significantly greater

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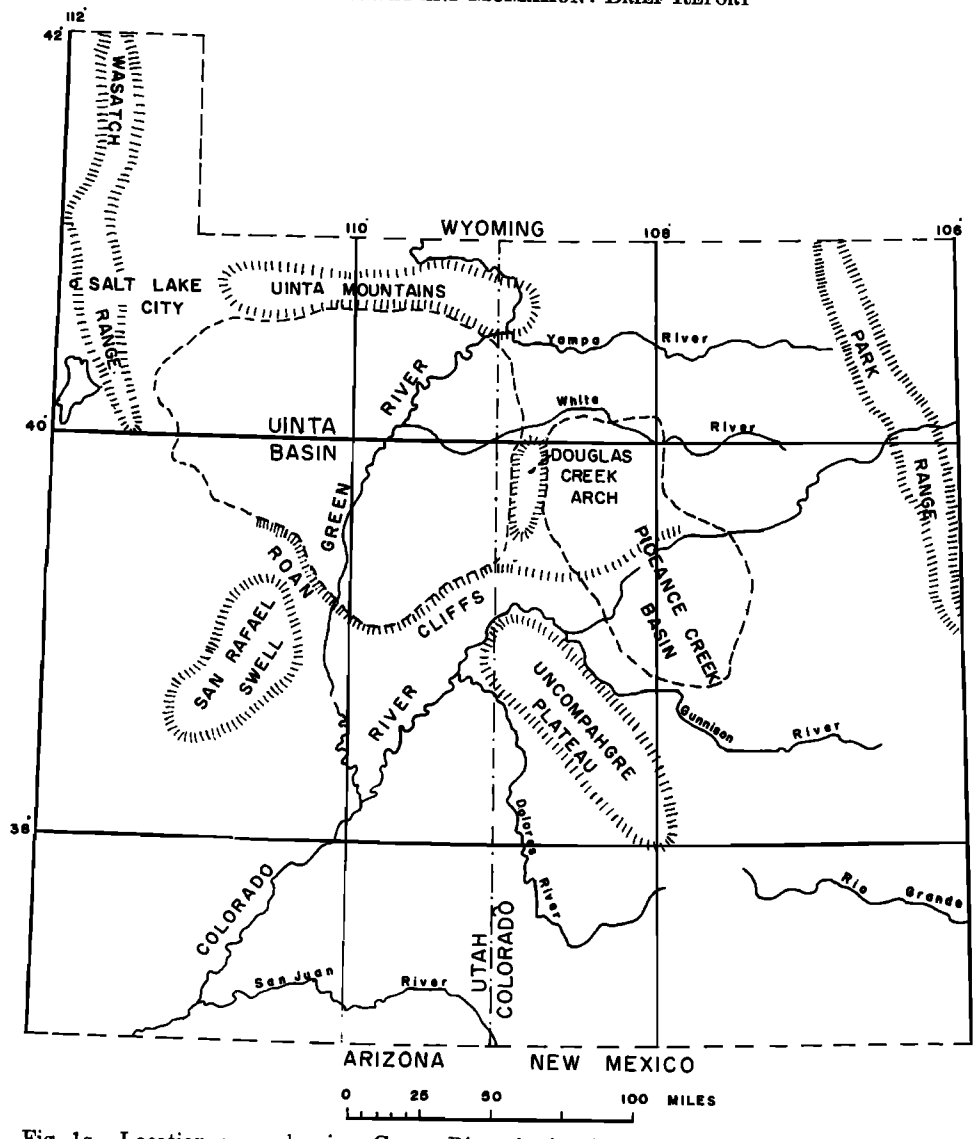


Fig. 1*a*. Location map showing Green River basins in Colorado and Utah. Samples are from Piceance Creek basin.

than 100 years can be recognized and that only features lasting several hundred years can be recognized with certainty. The total thickness of the sequence investigated by us, in detail, represents a time span of about 10,000 years. We should be able, therefore, to examine features of the field with periods of a few thousands of years.

Magnetic Properties

In an attempt to determine what magnetic phases were present in the Green River shale, we have looked at Curie temperatures and at magnetization versus magnetic field data (Figures 2*a* and 2*b*). The Curie temperature was run on a magnetic concentrate derived by crushing up some of the material and separating with a magnet. The Curie temperature is 580°C, indicating the presence of a small amount of magnetite. The measurements were done in air, and the cooling curve shows a small decrease in intensity, indicating a slight oxidation of the magnetite. The intensity of magnetization versus field up to 8 kG is almost a perfect straight line (Figure 2*b*). This allows us to make an upper estimate of the magnetite content, which,

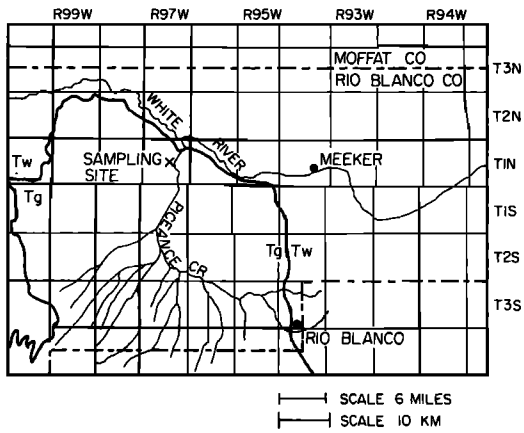


Fig. 1*b*. Local map showing sample location. Tg, Green River formation; Tw, Wasatch formation.

since it is ferromagnetic, would give a departure from the straight-line curve. There is, by this estimate, less than 0.002% magnetite present, indicating that there is very little magnetic material present, and the remanence can be expected to be very weak. The paramagnetic susceptibility is 4.8×10^{-6} emu/g, which is equivalent to 2.2% Fe in paramagnetic form (on the assumption that the paramagnetism is all due to Fe^{++}).

Measurement of Natural Remanent Magnetization

The measurement of natural remanent magnetization (NRM) in this sequence of rocks represents a major effort in which it was possible to average less than one useful measurement per day. The reason for this is that the samples are weakly magnetized, and, in fact, this study represents one of the very few successful attempts to work with such rocks. To illustrate some of the precautions taken to ensure that good measurements were obtained, we tabulate some of the experimental procedures.

1. The spinner magnetometer (PAR SM-2D) was very carefully aligned once a month.

2. Sample holders were made up, chemically and ultrasonically cleaned at frequent intervals, and used for these samples only. Typical volcanic samples, for example, invariably contaminated the sample holders. The sample holders were stored in a cabinet, since atmospheric particles often proved to be a problem.

3. Sample cups, sample holders, and the upper parts of the drive shaft were demagnetized in alternating fields at frequent intervals.

4. A single, very weak sample from the Green River suite was run every day as a check on the reliability of the results. It was found that measurements were not possible on all days, perhaps because of nearby electrical storms.

5. Blank measurements were made on the system at frequent intervals to be sure that the sample cup, sample holder, and shaft had not acquired any magnetization comparable to that of the sample levels.

6. In order to ensure that there was no

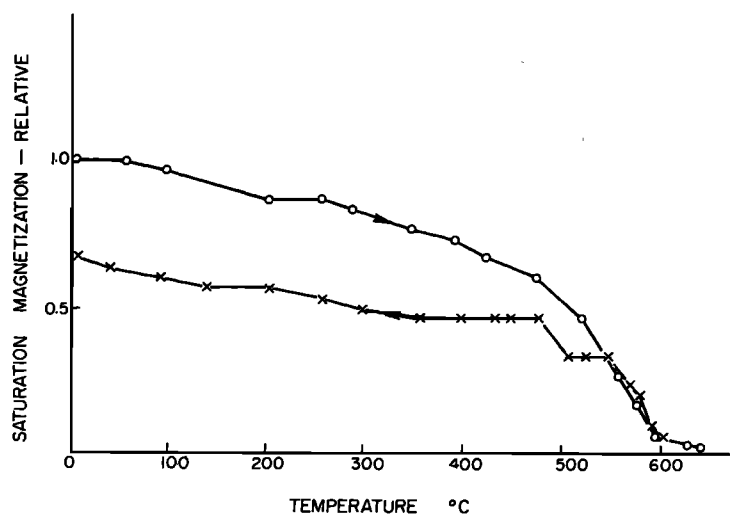


Fig. 2*a*. Curie temperature curve showing the presence of magnetite.

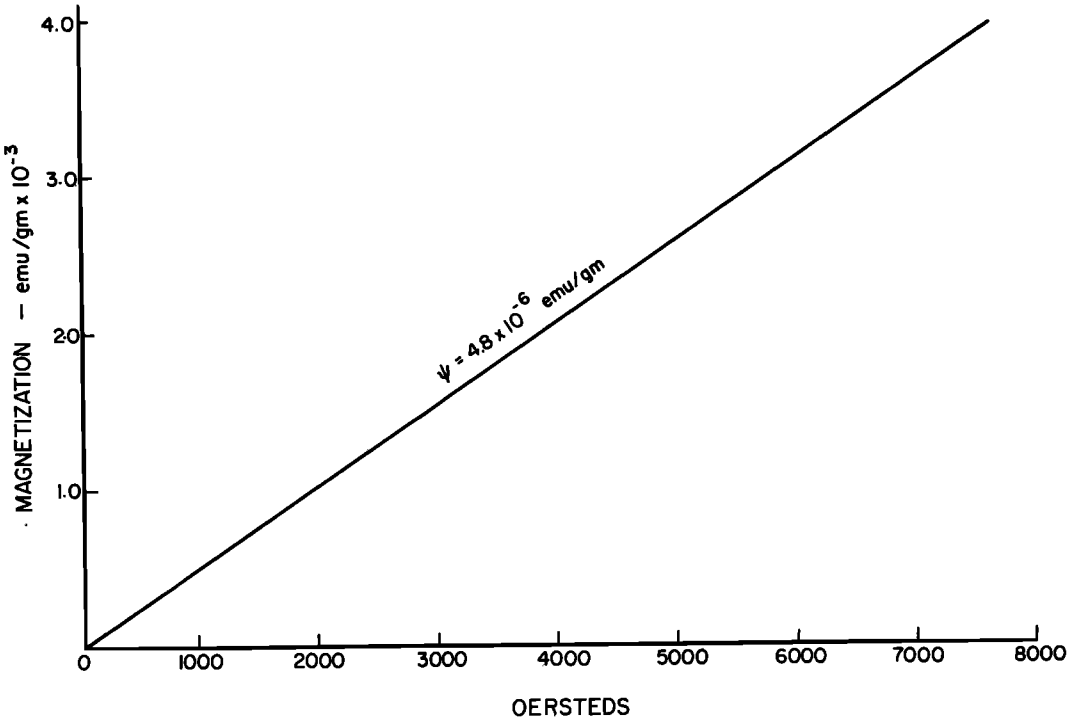


Fig. 2*b*. Magnetization versus field curve showing the presence of paramagnetism (straight-line continuous recording). Any ferromagnetic effects are below the level of detection (see text).

contamination on the samples from drilling or cutting, we ground the surface layer off about half of the samples with a diamond-tipped cutter in the lathe. This did not significantly change the results, and the procedure was eventually abandoned.

7. Since static charge was found to be a significant problem, we sprayed water mist on the spinning system in the manner suggested by *Graham* [1968].

8. It was not always possible to get closures from the redundant measurements by using only three spins. We therefore always used at least six spins (12 components) and averaged the data thus obtained to determine the mean intensity, declination, and inclination.

9. On several occasions, repeat measurements were made, and, where the observation was not repeatable to within about 5° or 10°, the samples were rejected.

We are convinced that these measurements are meaningful and represent true remanence in the samples. Since the time at which this work was performed, we have successfully worked with samples that are weaker by about a factor of 2 or 3, and so we know that the PAR spinner magnetometers, with proper care and feeding, can be used at levels of magnetization approaching 5×10^{-9} emu/cm^3, the quoted specifications.

Natural Remanent Magnetism

The NRM was measured on all samples. The intensity values are shown in Figure 3, which indicates that most samples have a mean intensity of about 5×10^{-8} to 10^{-7} emu/cm^3. These are very weak values, but, as indicated above, they are measurable. A few samples were chosen to conduct alternating field demagnetization tests. Figure 4 shows the results of some of these tests. In general, the intensity decreased fairly rapidly but monotonously, and the directions changed only slightly in most cases. At values of 150 and 200 oe the intensities became very weak, and we selected 100 oe as the optimum field for demagnetization. The intensities

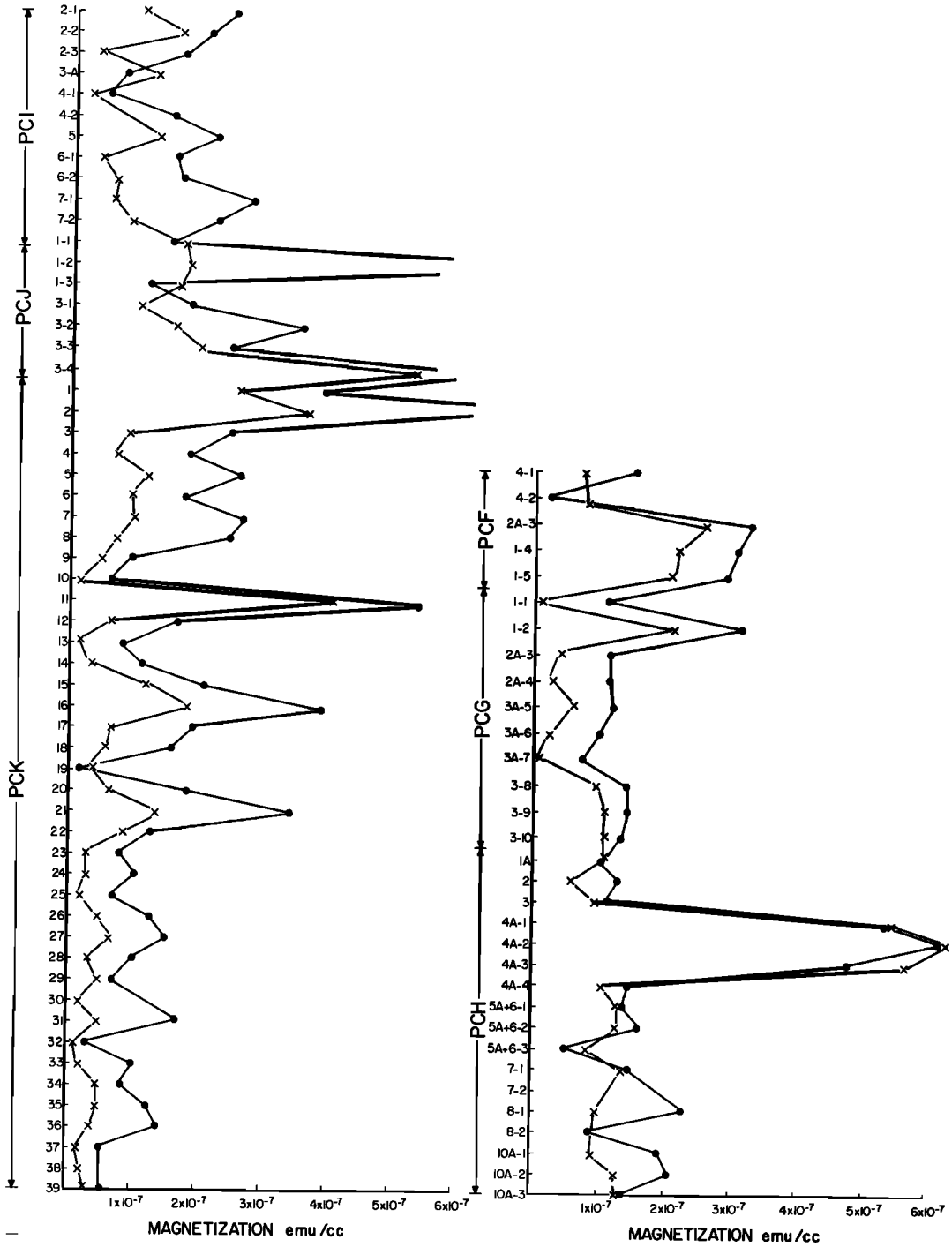


Fig. 3. Intensity of magnetization as a function of position in the sequence. Solid circles, before af demagnetization; crosses, after af demagnetization at 100 oe.

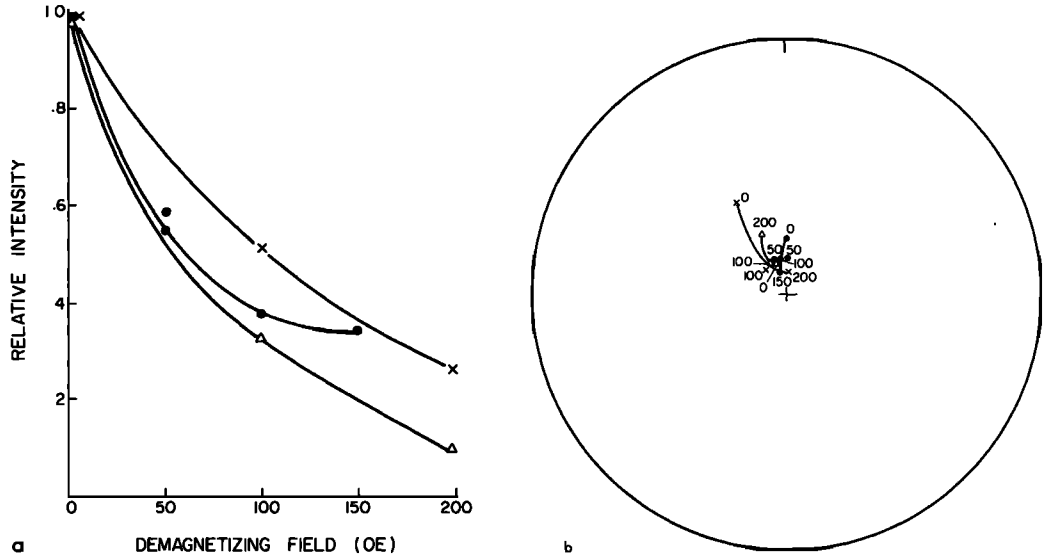


Fig. 4. Alternating field demagnetization plots of representative samples. Solid circles, PCK-7; crosses, PCK-25; open triangles, PCK-37. (*a*) Relative intensity as a function of the applied field. (*b*) Changes in direction on demagnetization (numbers refer to applied field).

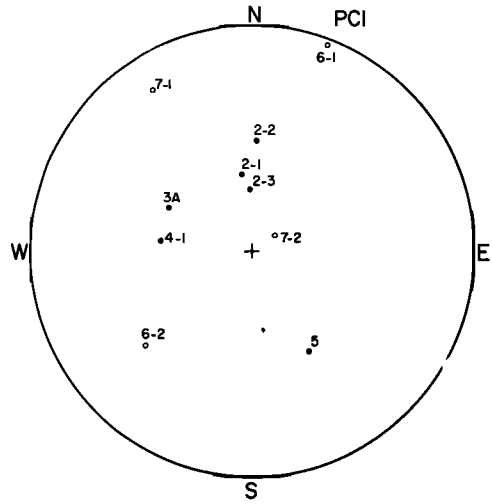
obtained in this way are also shown in Figure 3, which indicates values often as low as 1×10^{-8} emu/cm^3.

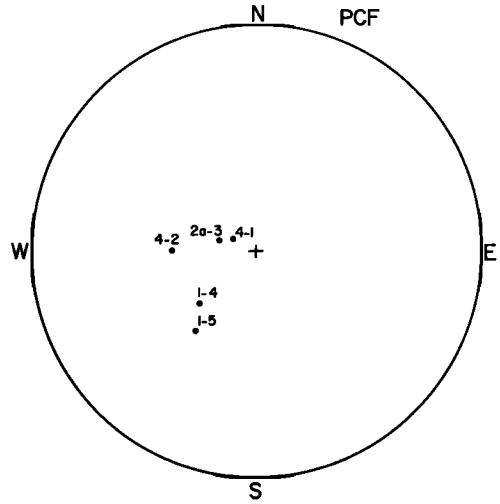
In spite of the various measuring difficulties, well-grouped sets of magnetic data (after cleaning) were obtained. These results are shown in Figure 5, arranged by group ranging from PCK at the bottom of the section to PCF at the top. The first obvious feature of these data is that there is a general grouping of directions, particularly for the PCK, PCJ, PCG, and PCF sets of data, that is normally magnetized (i.e., down) but has a westerly declination well away from the present field direction. In fact, this direction of magnetization is unusual in that it is well away from the direction normally found for Tertiary rocks in North America.

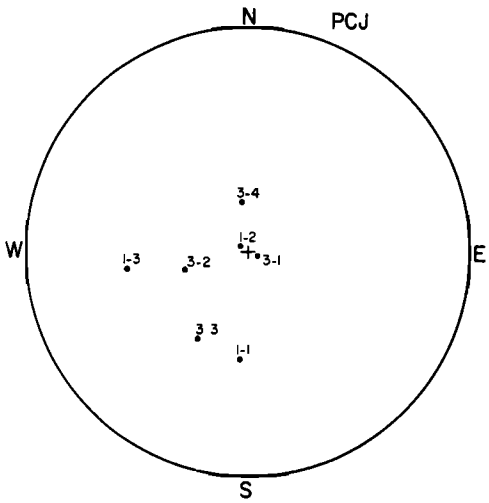
The subdivision of the data into the groups shown as PCK, PCJ, PCI, PCH, PCG, and PCF represents a natural grouping of the samples in stratigraphic sequence from F at the top to K at the bottom. The mean directions of magnetization (after cleaning to 100 oe) for samples in groups PCF, PCG, PCJ, and PCK are given in Table 1. The individual inclinations are shown in Figure 6 as a function of stratigraphic position, emphasizing the generally steep inclination. We have not shown the declination in the same plot because the dip is so steep, but the mean value of declination is 234°, a most unusual direction for rocks of this age. The overall mean pole position for these four groups of samples is also given in Table 1 and gives a mean position of 135°W, 19°N. This is quite unexpected for rocks of Tertiary age in North America and suggests that the mean pole position is the result of a deviation between the rotational pole and the mean magnetic field, at least on a local basis. This marked digression appears to have lasted for some thousands of years, suggesting that long-lived features of the earth's field of the type discussed by *Doell and Cox* [1971] are characteristic of the earth's magnetic field. They have shown that low secular variation is representative of the Pacific Ocean area over the last 0.7 m.y. It is also interesting to note that Eocene data collected by *Cox* [1957] from a series of volcanics in Oregon suggest a very unusual pole position.

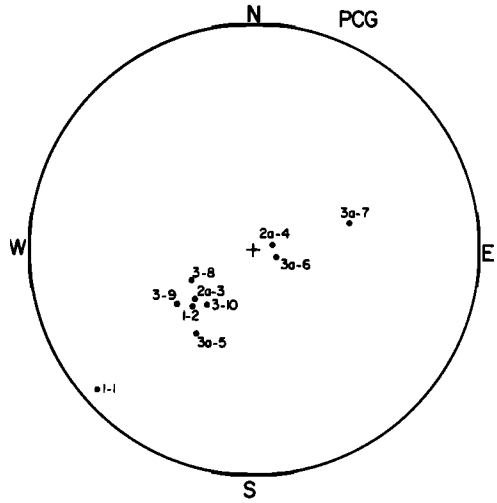
Data from the PCH and PCI series have not been included in the above discussion on mean

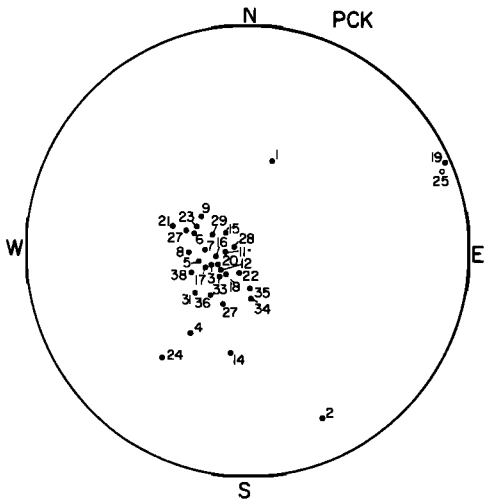
Fig. 5. (Opposite) Stereographic plot of directions of magnetization after cleaning in alternating fields of 100 oe. Stratigraphic sequence from PCF at top to PCK at bottom and in numerical sequence within each plot.

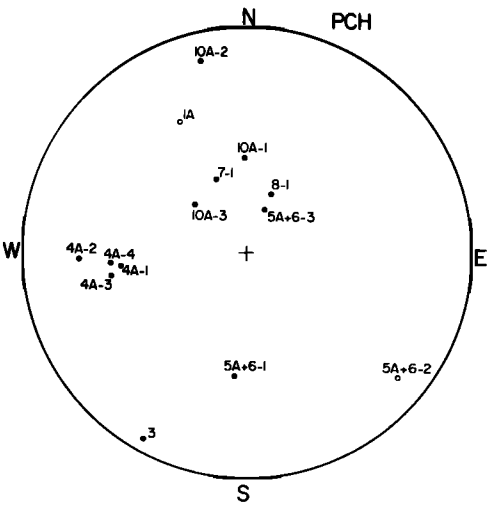

N
PCI
W
E
S
6-1
7-1
2-2
2-1
2-3
3A
4-1
7-2
6-2
5


N
PCF
W
E
S
4-2
2a-3
4-1
1-4
1-5


N
PCJ
W
E
S
3-4
1-2
1-3
3-2
3-1
3 3
1-1


N
PCG
W
E
S
3a-7
2a-4
3a-6
3-8
3-9
2a-3
3-10
1-2
3a-5
1-1


N
PCK
W
E
S
1
19
25
21
23
9
29
15
27
5
16
28
8
7
11
5
20
12
38
17
3
22
33
18
35
31
36
27
34
4
24
14
2


N
PCH
W
E
S
10A-2
1A
10A-1
7-1
8-1
10A-3
5A+6-3
4A-2
4A-4
4A-1
4A-3
5A+6-1
5A+6-2
3

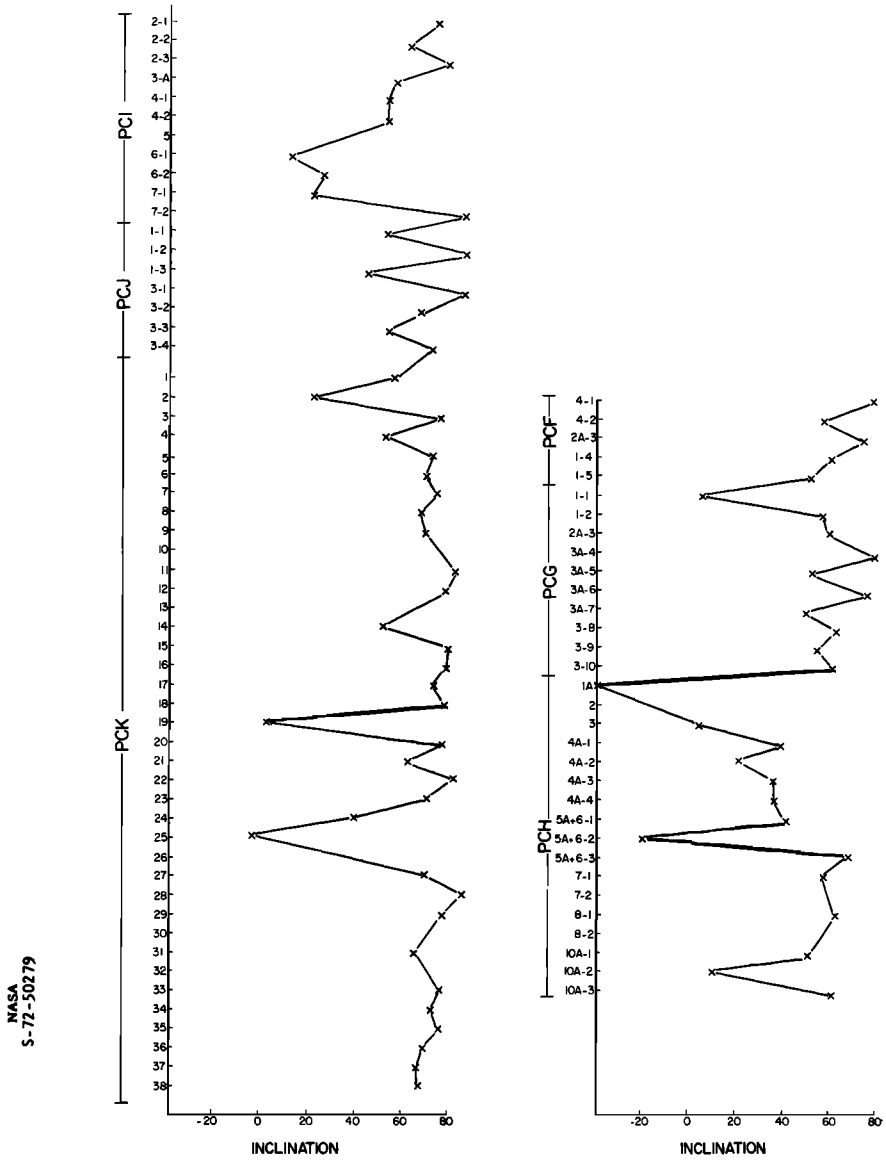


Fig. 6. Inclination (after af cleaning at 100 oe) as a function of position in the sequence.

TABLE 1. Summary of Paleomagnetic Data

Group	No. of Samples	D, deg	I, deg	k	α_{95}, deg	Longitude	Latitude	δM	δP
PCF	5	247	70	22.4	16.5	144	21	28	24
PCG	10	220	69	7.1	19.5	132	10	33	28
PCJ	7	235	74	9.8	20.3	134	21	37	33
PCK	33	233	77	8.2	9.3	129	25	17	16
Overall mean*	4	234	73	24.9	5.8	135	19	10	9

*Excludes PCH and PCI.

pole position, simply because these directions are highly scattered. Individual samples give repeatable or nearly repeatable measurements, but the variations from sample to sample are considerable and range from normal to reverse (Figure 5). Some of these results may be due to measuring or sampling difficulties, and we cannot be certain that these measurements represent ancient magnetic field conditions. Additional cores from the same samples show similar results, however. It is therefore at least possible that the earth's field was quite variable during this short period lasting for only two or three thousand years.

CONCLUSIONS

Although the Green River samples used in this study are very weakly magnetized, it is possible to get useful paleomagnetic results. The mean direction of magnetization is far enough from the apparent location of the north pole during the Eocene to suggest that there are long-lived deviations from the earth's dipolar field, at least on a local basis, with lifetimes of a few thousands of years or more.

Acknowledgments. The laboratory work was ably carried out by D. Redman and W. Mottley. We acknowledge assistance in sample collection from George Dana of the U.S. Bureau of Mines.

This work was supported by the National Research Council of Canada.

REFERENCES

Bradley, W. H., The varves and climate of the Green River epoch, *U.S. Geol. Surv. Prof. Pap., 158*, 87–110, 1928.

Bradley, W. H., Non-glacial marine varves, *Amer. J. Sci., 222*, 318–330, 1931.

Bradley, W. H., Geology of the Green River formation and associated Eocene rocks in southwestern Wyoming and adjacent parts of Colorado and Utah, *U.S. Geol. Surv. Prof. Pap., 496-A*, 1964.

Bradley, W. H., and H. P. Eugster, Geochemistry and paleolimnology of the Trona deposits and associated authigenic minerals of the Green River formation of Wyoming, *U.S. Geol. Surv. Prof. Pap., 496-B*, 1969.

Cashion, W. B., Geology and fuel resources of the Green River formation–southeastern Uinta basin, Utah and Colorado, *U.S. Geol. Surv. Prof. Pap., 548*, 1967.

Cox, A., Remanent magnetism of lower to middle Eocene basalt flows from Oregon, *Nature, 179*, 685–686, 1957.

Doell, R. R., and A. Cox, Pacific geomagnetic secular variation, *Science, 171*, 248–254, 1971.

Donnell, W. R., Tertiary geology and oil-shale resources of the Piceance Creek basin between the Colorado and White rivers, northwestern Colorado, *U.S. Geol. Surv. Bull., 1082-L*, 1961.

Evernden, J. F., D. E. Savage, G. H. Curtis, and G. T. James, Potassium-argon dates and the Cenozoic mammalian chronology of North America, *Amer. J. Sci., 262*, 145–198, 1964.

Graham, J. W., Elimination of static charges from spinner magnetometers, *J. Geophys. Res., 73*, 4788, 1968.

(Received October 26, 1972;
revised March 19, 1973.)